

Zhuoyuan Li

Current Location: West Lafayette, IN, USA | [GitHub: zhuoyl03](#)

li5917@purdue.edu | +1 (949) 678 3142

Education

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- Rensselaer Polytechnic Institute** — B.S. in Computer Science (Transferred) Aug 2023 – May 2025
- **Relevant Coursework:** Data Structures, Foundations of Computer Science, Computer Organization
 - **Honors:** Dean's Honor List (2023–2024)
- Purdue University, West Lafayette** — B.S. in Computer Engineering Technology Aug 2025 – May 2027
- GPA: 4.0/4.0
 - Expected Graduation: May 2027
 - **Relevant Coursework:** Linear Algebra, Probability, Introduction to Statistics

Research

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- LEO Satellite Network Resilience Analysis: A Focus on Critical Satellites** Sept 2024
- Zhuoyuan Li, Wenhao Chen, Yiyang Hu, Weyl Lu, Wenyi Zhang*
- Presented at the 2024 ACM Workshop on LEO Networking and Communication, co-located with MobiCom 2024 (LEO-NET)
- Performed a systematic resilience analysis of large-scale LEO satellite networks using the Hypatia simulator. Defined and identified *critical satellites* based on traffic usage frequency under ISL routing, and quantified their impact on end-to-end performance through targeted failure simulations.

Projects

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- Credit Card Fraud Detection System (MLOps)** [GitHub](#)
- Built an end-to-end machine learning system for fraud detection, covering training, online inference, monitoring, and drift detection.
 - Trained an XGBoost model on highly imbalanced transaction data with chronological splits and AUC-PR-based early stopping.
 - Designed a FastAPI service supporting batch scoring, rule-based decisions, health checks, and runtime metrics (latency, errors).
 - Implemented feature and score drift detection using PSI with sliding windows and severity classification.
 - Ensured reproducibility by versioning model artifacts, feature contracts, and training metadata.
 - Tools: Python, XGBoost, FastAPI, NumPy, Pandas, Conda
- RPI Campus Navigation System**
- Designed and implemented a modular campus navigation application for Rensselaer Polytechnic Institute using Java and JavaFX.
 - Modeled campus buildings and walkable connections as software abstractions, enabling flexible representation of map data and navigation logic.
 - Implemented shortest-path navigation functionality to compute and display optimal routes between buildings on the campus map.
 - Developed an event-driven graphical interface allowing users to select destinations either from a building list or by directly interacting with the map.
 - Enforced separation of concerns between UI components, navigation logic, and data models to improve maintainability and extensibility.
 - Tools Used: Java, JavaFX, Object-Oriented Design, Event-Driven Programming

Technologies

Programming: Python, C++ , C, Java, R, MATLAB

Machine Learning: Supervised Learning, Tree-based Models, Probabilistic Models

ML Systems: Model Training Pipelines, Online Inference, Monitoring, Data Drift Detection

Tools & Frameworks: XGBoost, scikit-learn, FastAPI, Pandas, NumPy